Where flux theory fails in the plane without rotation: a test with the chaos criterion of the literature

Willian Wistuba Melo da Cunha
Self-taught physics enthusiast

We test the three-body flux theory, with the measured absorptivity, in planar free falls with zero total angular momentum, requiring three to four scramble states before the escape, as in the literature, and the same threshold in absorption and in emission. At 95% confidence and with Holm correction, the theory passes in the mass sets 3-3-1 and 1-2-3, in the escape fractions and in the final eccentricity of the pair, but fails in the final eccentricity of the set 1-1-3. There, the failure grows with the depth of chaos: the Kolmogorov-Smirnov statistic goes from 0.082 to 0.126, and the measured mean eccentricity drops from 0.758 to 0.735, against 0.784 and 0.779 predicted, because of an excess of pairs of moderate eccentricity in the channel in which the pair of masses 3 and 1 remains. The failure does not come from the edges of the absorptivity grid, which we also measured for nearly radial pairs, or from the chaos criterion: an alternative criterion, based on democratic configurations, gives the same pattern. Rerunning the escapes with the geometry recorded, we show that only in 1-1-3 does the regime of the last episode strongly change the final shape of the pair: the mean eccentricity goes from 0.70, in the most mixed episodes, to 0.87, in the most hierarchical ones. Two explanations (more hierarchical time and more memory than in 1-2-3) were tested and ruled out. A replica with new seeds confirms the failure. A mass ladder 1-1-M, from M = 2 to 5, shows that it is not a narrow peak: with Holm correction, only the replica of M = 3 fails in the full shape, the neighbors M = 2.75 and 3.25 pass, and the measured mean eccentricity lies slightly below the predicted one at several steps, a pattern that is still exploratory. The mechanism remains open.

Keywords: three-body problem; flux theory; chaotic absorptivity; ergodicity; planar problem; zero angular momentum.

ORCID: 0009-0008-2875-6607

Contact: profwistuba01@gmail.com

Published: 5 November 2026. Version 1. Not peer reviewed.

Series: The three-body problem, part 15 of 15. https://fisica-revisitada.vercel.app/en/articles/onde-a-teoria-falha/

Translated from the original in Brazilian Portuguese.

I. INTRODUCTION

In Kol's flux theory, the probability of each outcome of a three-body system is the product of the phase-volume flux, known in closed form, and an absorptivity, which must be measured by simulation [1–3]. The theory holds for escapes that come from chaotic motion, and for this reason the tests in the literature only consider escapes preceded by at least three or four scramble states [2, 4]. A recent test in the planar case found discrepancies and raised the suspicion that, in the plane, chaos does not fully mix phase space [5]. That test used high total angular momentum and left the low-rotation regime open.

Notes FR-TC-13 and FR-TC-14 tested the theory in the plane with zero total angular momentum, but with a milder chaos criterion, of two scrambles [6, 7]. This note repeats the test with the criterion of the literature, examines two technical causes for the failure found, and looks for its mechanism in the geometry of the last chaotic episode.

II. METHOD

The simulations are those of the previous notes: planar, with zero total angular momentum and the chain integrator with algorithmic regularization [8]. We use three data sets.

- **Free falls:** 12,000 per mass set (1-1-3, 3-3-1, 1-2-3, and the control 1-1-1), starting from rest, with the number of scrambles of each episode and the eccentricity of the final pair [6].
- **Measured absorptivity:** 26,400 runs per channel (132 cells of relative energy $x = E_B/E_0$ and eccentricity e up to 0.995, with 200 runs each) and another 15,400 per channel at the edge of the nearly radial pairs (77 cells with $j = \sqrt{1-e^2}$ from 0.09 to 0.005, that is, e from 0.996 to 0.99999), counting the scrambles before the first excursion.
- **Geometry:** the escapes with deep chaos rerun from the same seeds, recording at each step the most bound pair and the position of the third body.
- **Replica and mass ladder:** 12,000 new free falls of 1-1-3, with other seeds, and a ladder 1-1-M with M = 2, 2.5, 2.75, 3.25, 3.5, 4, and 5, each step with 12,000 free falls, its own absorptivity measured in both channels (including the edge of nearly radial pairs), and the geometry of the last episode.

We use two chaos criteria. The first counts **scrambles**, stretches in which no pair has the third body beyond twice its semimajor axis. The second counts **democratic configurations**, the peaks of the homology radius above 0.33, which is the definition of Stone and Leigh [4]. In our data, the two go together (correlation of 0.90 to 0.95), and there are on average three democratic configurations per scramble.

The test is consistent: the same threshold k applies on both sides. In absorption, we count at least k events before the first excursion; in emission, at least k in the last episode before the escape. The prediction is the planar flux, with weight $x^{-2}e/\sqrt{1-e^2}$, times the measured absorptivity, interpolated in $(\ln x, j)$ [5, 6]. In this variable, the weight of the flux in eccentricity is uniform, which gives the edge $e \to 1$ the same resolution as the rest of the grid. The final eccentricity is compared by the Kolmogorov-Smirnov (KS) test, and the escape fractions by the χ^2 test. The critical value of the KS comes from 10,000 resamplings of the measured absorptivity, so that it includes the uncertainty of the prediction itself. The Holm correction is applied to the six comparisons of each family (full shape, shape up to $e = 0.995$, and escape fractions) [9]. In the ladder, the correction covers the 16 tests with new data in each family (seven steps and the replica, with k = 3 and 4), with 500 resamples. The geometry and sensitivity analyses are exploratory; the twin test was confirmed with a new sample, separate from the one used in the exploration.

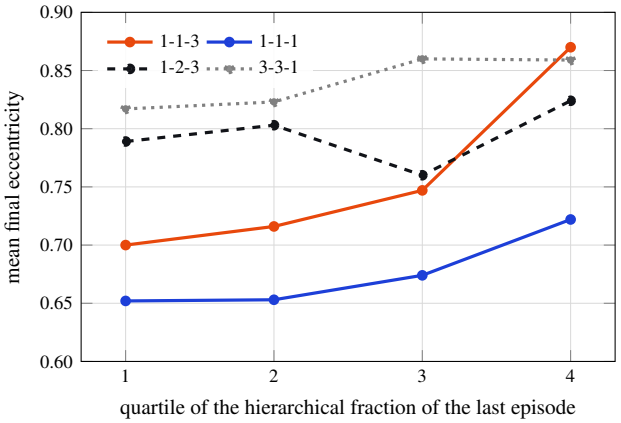


Fig. 1. Mean final eccentricity by quartile of the hierarchical fraction of the last episode (1 = least hierarchical time, 4 = most), in the escapes with at least three scrambles in the last episode. Only in 1-1-3 does the regime of the last episode change the shape of the final pair substantially.

III. RESULTS

A. The theory with the criterion of the literature

With three and four scrambles, the theory passes both questions in the sets 3-3-1 and 1-2-3 and fails in the final eccentricity of 1-1-3 (Table I). With two scrambles, 3-3-1 failed [7]. That failure disappears with deeper chaos, as expected from a short-range memory. The failure of 1-1-3, by contrast, grows: the KS goes from 0.082 to 0.126, and the measured mean eccentricity drops from 0.758 to 0.735, against 0.784 and 0.779 predicted. The deviation also appears when the test is restricted to the pairs with $e \leq 0.995$, and therefore it does not come from the edge of the nearly radial pairs. The escape fractions pass in all cases (p from 0.13 to 0.75 before the correction).

B. Two technical causes ruled out

Edges of the absorptivity grid. In energy, the absorptivity was measured up to $x = 20$, and above that we use the last measured value. Only 0.2% to 0.6% of the predicted escapes fall in this range, and replacing the absorptivity there with zero or with twice its value changes the KS of 1-1-3 by at most 0.002. In eccentricity, the edge weighs more: the planar flux puts about 10% of the weight above $e = 0.995$. For this reason we measured this range directly. The absorptivity there is lower than that of the last cell of the original grid, and the predicted fraction of pairs above 0.995 in 1-1-3 with four scrambles is 7.6%, close to the 7.1% measured. The edge accounts for about a quarter of the difference in the mean eccentricity. The rest remains, including in the test restricted to $e \leq 0.995$.

Where the deviation lies. In 1-1-3, the escape of a light body, which leaves the pair of masses 3 and 1, accounts for 97% of the escapes, with the predicted frequency (97.0% against 97.3% with four scrambles). Within this channel, 62.8% of the pairs end with $e \leq 0.872$, against 49.9% predicted (548 escapes; with three scrambles, 58.3% against 50.2% in 1,153 escapes). There is an excess of pairs of moderate eccentricity within the same channel.

Chaos criterion. With the criterion of democratic configurations, the pattern repeats (Table II). With six configurations, shallow chaos, 3-3-1 fails and 1-1-3 passes. With eight and ten, equivalent to about three scrambles, 3-3-1 and 1-2-3 pass and 1-1-3 fails, with the measured mean eccentricity below the predicted one. The failure of 1-1-3 therefore does not depend on the chosen criterion.

C. Two regimes in the last episode

We reran the escapes with at least three scrambles in the last episode: 1,181 in 1-1-3, 1,159 in 1-2-3, 1,223 in 3-3-1, and 1,275 in 1-1-1. In all of them, the same body escaped and the final eccentricity matched the original one. We divided the time of the last episode into two types. In the **hierarchical** regime, the most bound pair is well defined and the third body lies beyond twice its semimajor axis. In the remaining stretches, the third body is close and the pairs alternate.

In 1-1-3, the final eccentricity depends strongly on the hierarchical fraction of the last episode: the Spearman correlation is +0.295 ($p = 4 \times 10^{-25}$), and the mean eccentricity goes from 0.70 in the quartile with the least hierarchical time to 0.87 in the quartile with the most (Fig. 1). In the other three sets, the relation is weak, between +0.06 and +0.10 (Table III). Pairs that end after a more mixed episode come out more circular. This is consistent with the drop in the mean eccentricity when deeper chaos is required.

A first hypothesis, that 1-1-3 would spend more time in the hierarchical regime, was registered before the measurement in the sets 1-1-1 and 3-3-1 and was not confirmed: 1-1-1 spends more time in the hierarchical regime (median of 0.55) and does not show the effect. What distinguishes 1-1-3 is not the

Table I. Consistent test of the flux theory with the measured absorptivity and the scramble criterion. KS of the final eccentricity, with the 95% critical value obtained by 10,000 resamplings, and p before the correction, for all escapes and only for those with $e \leq 0.995$. Measured and predicted mean eccentricity. With the Holm correction, only 1-1-3 fails, in both shape tests.

k	Set (n)	KS (critical)	p	KS up to 0.995 (critical)	p	mean e
3	1-1-3 (1,181)	0.082 (0.060)	0.004	0.101 (0.065)	0.0005	0.758 / 0.784
3	3-3-1 (1,223)	0.045 (0.064)	0.25	0.060 (0.071)	0.12	0.840 / 0.850
3	1-2-3 (1,159)	0.021 (0.058)	0.90	0.031 (0.063)	0.64	0.794 / 0.793
4	1-1-3 (565)	0.126 (0.088)	0.002	0.133 (0.093)	0.002	0.735 / 0.779
4	3-3-1 (604)	0.055 (0.093)	0.43	0.053 (0.104)	0.56	0.843 / 0.838
4	1-2-3 (552)	0.043 (0.084)	0.62	0.046 (0.091)	0.61	0.810 / 0.801

Table II. The same test with the criterion of democratic configurations (homology radius above 0.33), with 100 resamplings, without correction for multiple comparisons, and with the absorptivity grid up to $e = 0.995$, repeating the last value above that (exploratory). KS, with the 95% critical value in parentheses, and p; in brackets, the number of escapes.

k	1-1-3	3-3-1	1-2-3
6	0.031 (0.040), $p = 0.29$ [2,556]	0.085 (0.044), $p < 0.01$ [2,963]	0.018 (0.044), $p = 0.83$ [2,463]
8	0.085 (0.056), $p < 0.01$ [1,517]	0.038 (0.065), $p = 0.49$ [1,408]	0.026 (0.054), $p = 0.79$ [1,371]
10	0.113 (0.075), $p < 0.01$ [988]	0.057 (0.082), $p = 0.34$ [989]	0.038 (0.075), $p = 0.65$ [909]

proportion of each regime, but how differently the two regimes shape the resulting pairs.

D. Twin test

To measure the memory directly, we stopped each escape at the start of its last episode and launched 20 copies from there, with the positions perturbed by a fraction δ of the system scale. With $\delta = 10^{-6}$, the copies practically repeat the original (spread between copies of 0.06 to 0.09, against 0.20 to 0.23 in the whole set), because the episode is too short for chaos to amplify the perturbation, and the test is not informative. With $\delta = 10^{-2}$, in a new sample of 800 escapes per set, the correlation between the final eccentricity of the original and the mean of the copies with at least three scrambles is 0.31 in 1-1-3 (579 originals; 95% interval from 0.23 to 0.38) and 0.21 in 1-2-3 (589 originals; from 0.14 to 0.29). The difference is not significant ($p = 0.08$). The two sets retain similar memory, and 1-2-3 passes the test of the theory nonetheless, so this memory does not explain the failure of 1-1-3. An opposite result, seen in an exploration with 200 escapes, was not confirmed.

E. Replica and mass ladder

Replica. With 12,000 new free falls of 1-1-3, the failure repeats (Table IV). The KS is 0.101 with three scrambles and 0.126 with four, and the measured mean eccentricity is 0.750 and 0.738, against 0.784 and 0.779 predicted. In the channel in which the pair of masses 3 and 1 remains, 60.5% of the pairs end with $e \leq 0.872$, against 50.3% predicted. With three scrambles, the replica also fails in the fraction of escapes of the heavy body (3.95% measured against 2.55% predicted, $p = 0.003$), which did not happen in the original sample (2.37%). With the two samples combined, the fraction is 3.15%, and the difference is not significant ($p = 0.07$). With four scrambles, the escape fraction passes ($p = 0.38$).

Ladder. To find out whether the failure is a peak at $M = 3$, we repeated the test in 1-1-M with seven other values of the central mass (Table IV). With Holm correction, only the replica of $M = 3$ fails in the full shape, with k = 3 and 4, and, in the test restricted to $e \leq 0.995$, the replica and the step $M = 2.5$ with $k = 3$. The neighbors M = 2.75 and 3.25 pass, with a difference in mean eccentricity close to zero, so the failure is not a narrow resonance. The difference between the measured and predicted mean eccentricity is negative in most steps from 2.5 on, and its 95% interval excludes zero at $M = 2.5$ (from -0.047 to -0.012 with $k = 3$) and at $M = 5$ (from -0.054 to -0.007). There is no significant trend with mass: the slope against $\ln M$ has $p = 0.15$ with $k = 3$ and $p = 0.59$ with $k = 4$. This small, broad deficit was noticed after seeing the data and is exploratory.

Geometry on the ladder. The link between the hierarchical fraction of the last episode and the final eccentricity exists at every step and peaks near $M = 3$. The Spearman correlation is +0.12 at $M = 2$, +0.15 at 2.5, +0.21 at 2.75, +0.30 at 3, +0.26 at 3.25, +0.25 at 3.5, +0.20 at 4, and +0.16 at 5. It is also strong at M = 3.25 and 3.5, where the theory passes, so it is not enough to explain the failure.

IV. DISCUSSION

With the chaos criterion of the literature, the flux theory with the measured absorptivity describes the plane without total rotation in two of the three sets tested and fails robustly in 1-1-3. The failure does not come from the edges of the

Table III. Fraction of the last episode in the hierarchical regime (median and quartiles) and Spearman correlation between this fraction and the final eccentricity, in the escapes with at least three scrambles in the last episode.

Set (n)	Hierarchical fraction: median (quartiles)	Correlation with final e
1-1-1 (1,275)	0.55 (0.44 to 0.63)	+0.090 ($p = 0.001$)
1-1-3 (1,181)	0.41 (0.28 to 0.53)	+0.295 ($p = 4 \times 10^{-25}$)
1-2-3 (1,159)	0.25 (0.14 to 0.36)	+0.061 ($p = 0.04$)
3-3-1 (1,223)	0.16 (0.06 to 0.27)	+0.095 ($p = 0.001$)

Table IV. Coherent test on the ladder 1-1-M, with 500 resamples. KS of the final eccentricity, with the 95% critical value in parentheses; p before correction, with the p of the test restricted to $e \leq 0.995$ in parentheses (0.002 is the smallest possible value with 500 resamples); Δe is the measured mean eccentricity minus the predicted one. There are 1,138 to 1,279 escapes per step with $k = 3$ and 558 to 683 with $k = 4$.

M	KS, $k = 3$	p	Δe	KS, $k = 4$	p	Δe
2	0.053 (0.058)	0.092 (0.062)	+0.004	0.052 (0.085)	0.41 (0.37)	−0.006
2.5	0.075 (0.061)	0.016 (0.002)	−0.030	0.080 (0.088)	0.080 (0.042)	−0.028
2.75	0.036 (0.060)	0.43 (0.40)	−0.007	0.043 (0.089)	0.63 (0.66)	−0.008
3, original	0.081 (0.059)	0.006 (0.002)	−0.026	0.125 (0.086)	0.004 (0.004)	−0.044
3, replica	0.101 (0.060)	0.002 (0.002)	−0.035	0.126 (0.086)	0.002 (0.004)	−0.041
3.25	0.040 (0.060)	0.35 (0.47)	−0.018	0.035 (0.087)	0.83 (0.77)	−0.005
3.5	0.074 (0.066)	0.020 (0.006)	−0.019	0.085 (0.090)	0.078 (0.11)	−0.023
4	0.054 (0.060)	0.12 (0.060)	−0.019	0.041 (0.090)	0.73 (0.76)	+0.010
5	0.065 (0.066)	0.058 (0.072)	−0.030	0.082 (0.096)	0.11 (0.13)	−0.040

absorptivity grid or from the chaos criterion, it also appears also when only the pairs with $e \leq 0.995$ are considered alone, and it grows when deeper chaos is required, contrary to a short-range memory effect. This is compatible with the suspicion that, in the plane, the mixing of phase space may be incomplete [5], now in the zero-rotation regime, and with the coexistence of regular and chaotic regions described in the literature [10, 11].

The geometry indicates where to look. Only in 1-1-3 does the type of motion of the last episode change the final shape: more hierarchical episodes give eccentric pairs, and more mixed episodes give circular pairs. This is a correlation, not a proof of cause. The boundary between the regimes (twice the semimajor axis) is our choice. The mass ladder shows that this link peaks near $M = 3$, where the failure is largest, but remains strong at neighboring steps where the theory passes. The failure of 1-1-3, now replicated, seems to be the visible part of a small eccentricity deficit that appears at several steps; confirming this deficit requires new samples with the tolerance fixed beforehand.

Limitations:

- everything holds for planar simulations with zero total angular momentum and four mass sets;
- with four scrambles, 552 to 683 escapes remain per set;
- the ladder uses 500 resamples, and the broad eccentricity deficit in it is exploratory;
- the geometry analyses and Table II are exploratory;
- the scramble criterion is not identical to the one used by the reference groups, although the criterion of democratic configurations gives the same pattern.

V. CONCLUSIONS

- With three and four scrambles, the criterion of the literature, the flux theory with measured absorptivity passes, at 95% confidence and with Holm correction, in the sets 3-3-1 and 1-2-3, in the escape fractions and in the final eccentricity.
- In the set 1-1-3, it fails in the final eccentricity, and the failure grows with the depth of chaos (KS from 0.082 to 0.126; measured mean eccentricity from 0.758 to 0.735, against 0.784 and 0.779 predicted). The deviation is an excess of pairs of moderate eccentricity in the channel in which the pair of masses 3 and 1 remains.
- The failure does not come from the edges of the absorptivity grid, also measured for nearly radial pairs, or from the chaos criterion.
- Only in 1-1-3 does the regime of the last episode strongly change the final shape of the pair (from 0.70 to 0.87 between quartiles of hierarchical time). The hypotheses of "more hierarchical time" and of "more memory than 1-2-3" were tested and ruled out.
- A replica with new seeds confirms the failure in shape. On the ladder 1-1-M, from $M = 2$ to 5, the failure is not

a narrow peak: the neighbors 2.75 and 3.25 pass, and the measured eccentricity lies slightly below the predicted one at several steps, a pattern that is still exploratory.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

CITE AS

W. W. M. da Cunha, "Where flux theory fails in the plane without rotation: a test with the chaos criterion of the literature," Física Revisitada, Technical Note FR-TC-15 (2026), https://fisica-revisitada.vercel.app/en/articles/onde-a-teoria-falha/.

[1] B. Kol, Flux-based statistical prediction of three-body outcomes, Celest. Mech. Dyn. Astron. 133, 17 (2021). doi.org/10.1007/s10569-021-10015-x

[2] V. Manwadkar, B. Kol, A. A. Trani, and N. W. C. Leigh, Testing the flux-based statistical prediction of the three-body problem, Mon. Not. R. Astron. Soc. 506, 692 (2021). arxiv.org/abs/2101.03661

[3] V. Manwadkar, A. A. Trani, and B. Kol, Measurement of three-body chaotic absorptivity predicts chaotic outcome distribution, Celest. Mech. Dyn. Astron. 136, 4 (2024). doi.org/10.1007/s10569-023-10174-z

[4] N. C. Stone and N. W. C. Leigh, A statistical solution to the chaotic, non-hierarchical three-body problem, Nature 576, 406 (2019). doi.org/10.1038/s41586-019-1833-8

[5] Y. Dandekar, E. Springer, S. Zard, A. A. Trani, B. Kol, and O. Birnholtz, Planar three-body problem: theoretical predictions and simulation results, arXiv:2608.30414 (2026). arxiv.org/abs/2608.30414

[6] W. W. M. da Cunha, Física Revisitada, Technical Note FR-TC-13 (2026). fisica-revisitada.vercel.app/artigos/muro-de-excentricidade/

[7] W. W. M. da Cunha, Física Revisitada, Technical Note FR-TC-14 (2026). fisica-revisitada.vercel.app/artigos/formula-posta-a-prova/

[8] S. Mikkola and K. Tanikawa, Algorithmic regularization of the few-body problem, Mon. Not. R. Astron. Soc. 310, 745 (1999). doi.org/10.1046/j.1365-8711.1999.02982.x

[9] S. Holm, A simple sequentially rejective multiple test procedure, Scand. J. Stat. 6, 65 (1979). www.jstor.org/stable/4615733

[10] A. A. Trani, N. W. C. Leigh, T. C. N. Boekholt, and S. Portegies Zwart, Isles of regularity in a sea of chaos amid the gravitational three-body problem, arXiv:2403.03247 (2024). arxiv.org/abs/2403.03247

[11] H. D. Parischewsky, A. A. Trani, and N. W. C. Leigh, Chaos in the vicinity of a singularity in the three-body problem: the equilateral triangle experiment in the zero angular momentum limit, SciPost Phys. Core 6, 016 (2023). arxiv.org/abs/2108.06335